REPORT

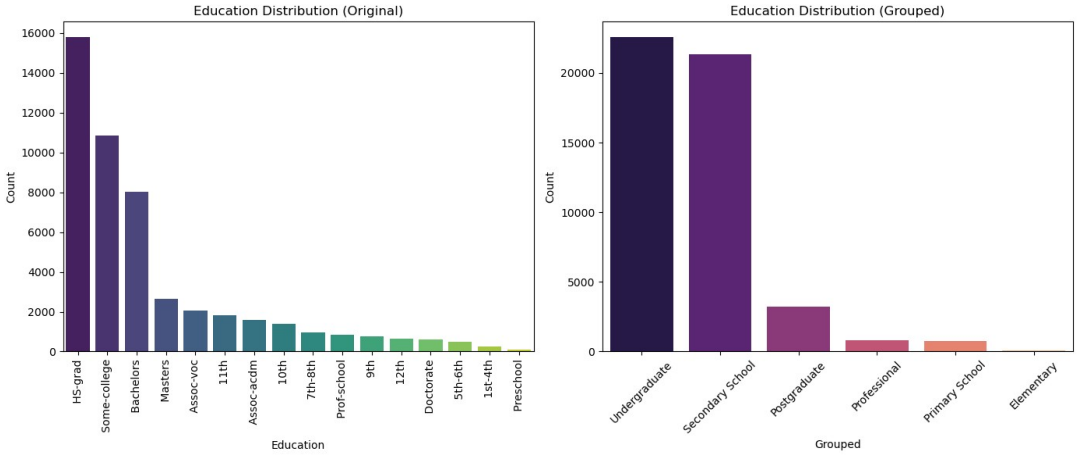
PART 1

Q1

The grouping was done to simplify the data into stages of schooling and higher studies:

- **Elementary**: Preschool – very beginning of formal education.
- **Primary School**: Grades 1–6 – basic foundational education.
- **Secondary School**: Grades 7–12 and High School graduates – covers middle and high school years.
- **Undergraduate**: Some-college, Associate degrees, and Bachelors – represents higher education before advanced degrees.
- **Postgraduate**: Masters and Doctorate – advanced academic study and research.
- **Professional**: Professional school degrees – career-focused programs such as medicine or law.

This grouping makes the progression of education clearer and easier to analyze.



Findings from the Grouped Distribution

The grouped chart shows that most individuals are concentrated in **Secondary School (~21,500)** and **Undergraduate (~22,500)** categories. This indicates that the majority of people either completed schooling or pursued some form of higher education at the undergraduate level.

The **Postgraduate (~3,200)** and **Professional (~1,000)** groups are much smaller, showing that fewer people progress to advanced degrees or specialized professional training.

The **Primary School (~900)** and **Elementary (~50)** groups are very small, suggesting that most individuals in the dataset continued their studies beyond the early stages of schooling.

Q2

Grouping of Variables

To simplify interpretation, both age and work hours were grouped into broader, more meaningful categories:

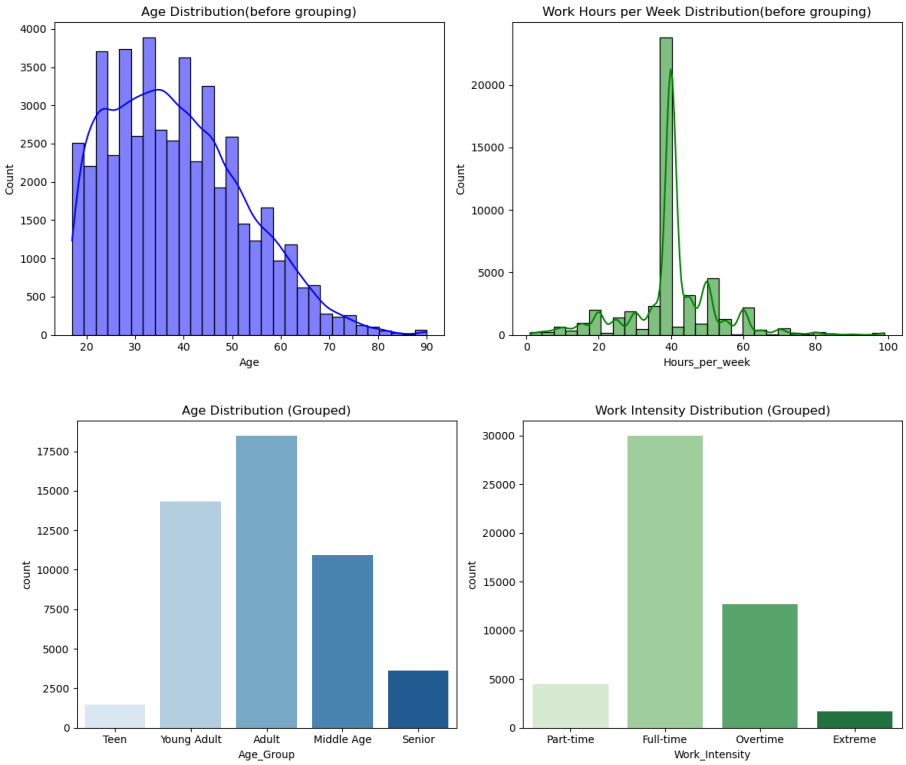
- **Age Groups**:
  - *Teen* (≤18)
  - *Young Adult* (19–30)
  - *Adult* (31–45)
  - *Middle Age* (46–60)
  - *Senior* (61+)
- **Work Intensity Groups**:
  - *Part-time* (≤20 hours)
  - *Full-time* (21–40 hours)
  - *Overtime* (41–60 hours)
  - *Extreme* (61+ hours)

This classification mirrors real-world life stages and common employment categories, making the analysis more intuitive than raw values.

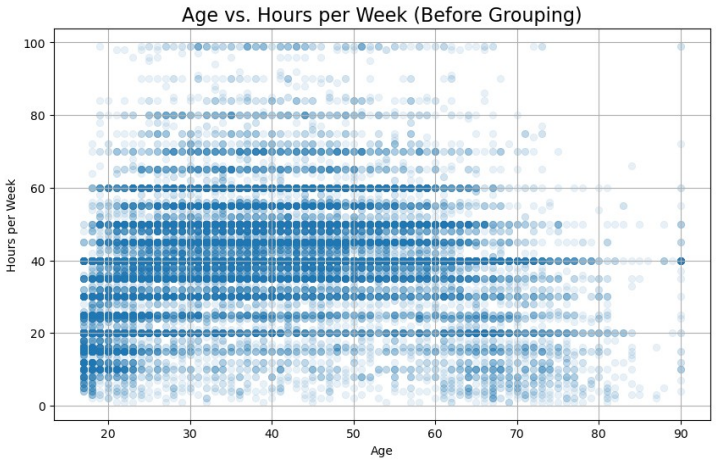
Distributions After Grouping

After applying the grouping, clearer demographic and employment patterns emerge:

- **Age**: The largest share of individuals fall into the *Adult* (31–45) group, followed by *Young Adults* (19–30). *Middle Age* workers (46–60) form a smaller group, while *Teens* and *Seniors* are least represented.



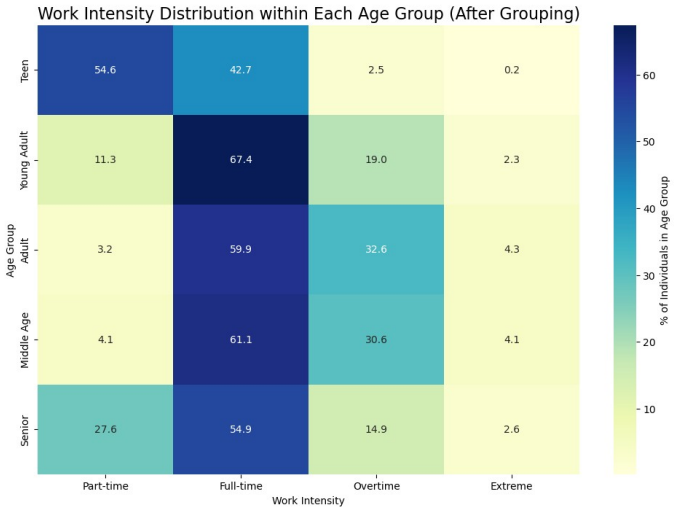
- **Work Intensity**: *Full-time* dominates with nearly 30,000 individuals, reflecting standard working hours. *Overtime* is the next most common, while *Part-time* and *Extreme* categories are relatively small minorities.



It is important to note that while hours worked ranged from very few to nearly 100, most individuals clustered around 40 hours, which explains why the *Full-time* category is overwhelmingly large.

Relationship Between Age and Work Intensity

The scatterplot initially suggested little to no relationship between age and hours worked, supported by a weak correlation coefficient of **0.0713**. However, after grouping, more meaningful age-specific patterns become visible:



- **Teens** are split between *Part-time* (54.6%) and *Full-time* (42.7%), consistent with students often working fewer hours.
- **Young Adults** show the strongest tendency for *Full-time* work (67.4%), with some moving into *Overtime* as careers develop.
- **Adults (31–45)** and **Middle Age (46–60)** continue this trend, with most in *Full-time* (~60%) and a notable share in *Overtime* (~30%).
- **Seniors** stand out, as more individuals shift to *Part-time* (27.6%), reflecting reduced workloads or semi-retirement.

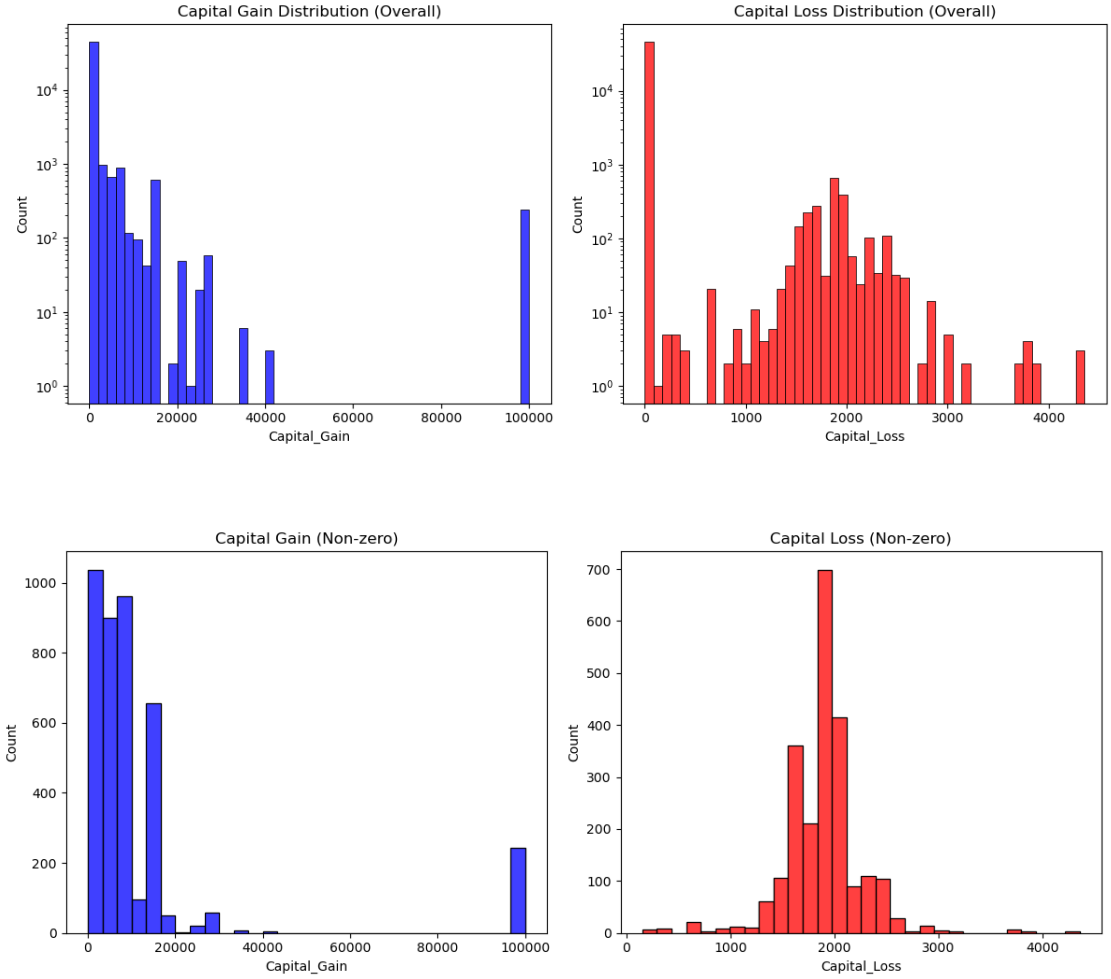
Thus, while raw data showed almost no correlation, grouped data highlights life-stage transitions in employment intensity.

Interpretability vs. Detail

The grouping greatly improves interpretability by condensing complex data into recognizable categories. It reveals patterns such as teens' tendency toward part-time work and seniors' gradual withdrawal from the workforce, which were hidden in raw scatterplots.

At the same time, this process reduces detail. For example, treating all "extreme" workers—whether 65 or 90 hours—as one group loses fine-grained variation. Similarly, both a 19-year-old and a 29-year-old are classified as *Young Adults*, despite possible differences in work behavior. This trade-off is typical when balancing simplicity and precision.

Q3



Distribution of Capital Gain and Loss

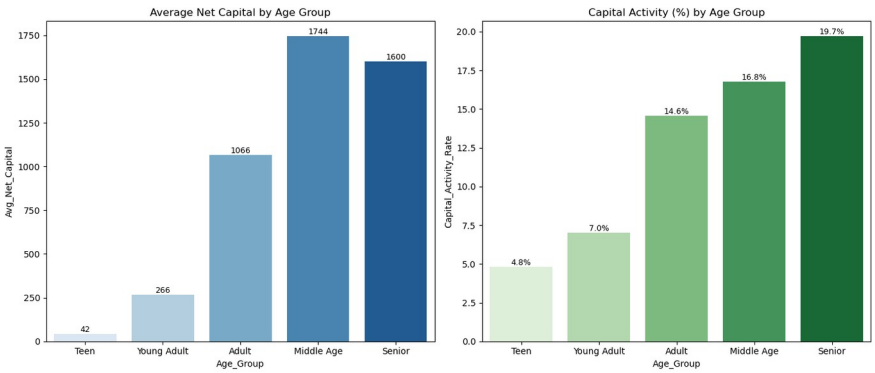
The distributions of capital gain and capital loss are highly skewed. Most individuals report zero capital gain or loss, with only a small portion participating in such financial activity.

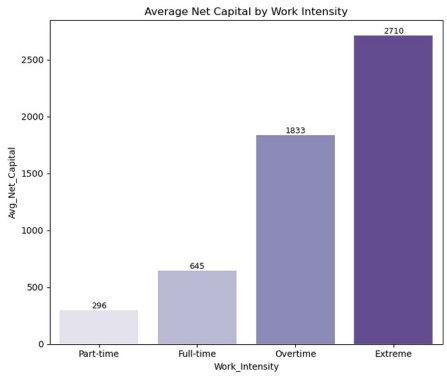
For those who do report gains, values are typically modest, though a few individuals report extremely high gains, suggesting the presence of income caps or outliers. Losses, while also sparse, appear more concentrated and symmetrical, indicating that those who incur losses do so within a more predictable range.

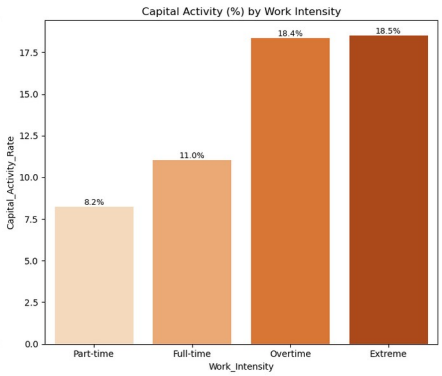
Among individuals with non-zero capital activity, capital gains still exhibit a strong right-skew, while capital losses appear more normally distributed, peaking at moderate values. This suggests that while high capital gains are possible, they are uncommon, whereas capital losses tend to cluster around a typical range.

Capital Activity Rate

The vast majority of individuals do not engage in capital gain or loss. Only a small fraction of the







population—less than one-fifth—reports any form of capital activity. This indicates that capital activity is not a widespread economic behavior, but rather limited to a specific subset of individuals.

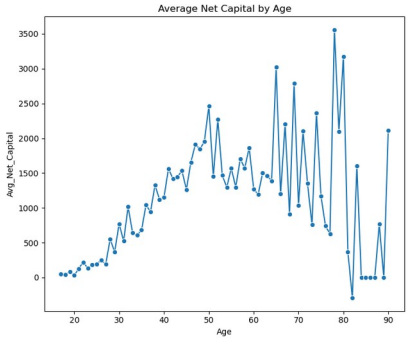
Age and Work Intensity Group Analysis

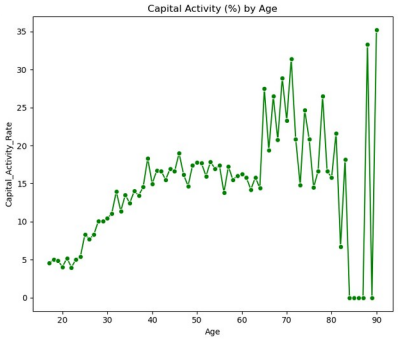
By Age Group

When grouping by age (e.g., teens, young adults, adults, middle-aged, and seniors), clear trends emerge. Both the average net capital and the likelihood of capital activity tend to increase with age. Middle-aged and senior individuals report significantly higher net capital compared to younger age groups. Additionally, older individuals are more likely to have engaged in some form of capital activity, reflecting accumulated wealth and greater financial involvement over time.

By Work Intensity

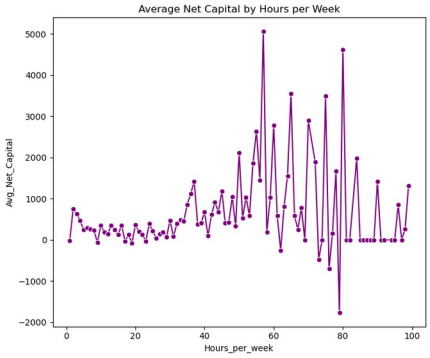
Grouping by work intensity (e.g., part-time, full-time, overtime, and extreme workloads) reveals an even stronger association. As work intensity increases, so does the average net capital. Individuals in high-intensity work categories not only accumulate more capital on average but are also more likely to report capital gains or losses. This suggests that work engagement is a key driver of financial activity beyond wages alone.

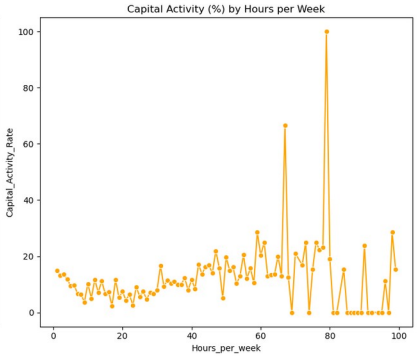




Trends Across Continuous Variables

When examining capital trends by continuous age and hours worked per week, the patterns largely reflect those seen in grouped data. Net capital tends to rise steadily with age up to a certain point before plateauing or declining in older age brackets. A similar upward trend is visible with increasing work hours, although variability becomes more pronounced at the extremes.





Capital activity rates also tend to rise with age and work hours, but exhibit greater volatility among older individuals and those working extremely high hours. These fluctuations may reflect smaller population sizes in those groups or diverse financial behaviors.

Discussion and Interpretation

Overall, both **age** and **work intensity** show positive relationships with capital outcomes, but **work intensity** appears to have a stronger and more consistent association with both net capital and the likelihood of capital activity. While age reflects life-stage accumulation of assets, work intensity is more directly tied to income-generating behavior and access to capital investment opportunities.

Grouping individuals into categories like age groups or work intensity levels simplifies interpretation and reveals broader patterns. However, it may obscure nuanced trends that are more evident in continuous data. Thus, combining both grouped and continuous analyses offers a more comprehensive view of capital behavior.

Q4

Missing Value Handling

- **Categorical columns** (`Workclass`, `Education`, `Occupation`, etc.) were filled using the **mode**.
- **Numerical columns** (`Capital_Gain`, `Capital_Loss`, `Hours_per_week`, etc.) were filled using the **median**.
- Result: **All missing values** were resolved.

Duplicate Removal

- Duplicate rows were identified and **removed**, reducing redundancy and improving data quality.

Feature Transformations

- **New feature**: `Net_Capital` = `Capital_Gain - Capital_Loss`.
- **Grouped features**:
  - `Age_Group`: Age bucketed into 5 life-stage categories.
  - `Work_Intensity`: Derived from weekly hours into 4 categories (e.g., Part-time, Extreme).
- **New indicator**: `Capital_Activity_Rate` to show any gain/loss activity.

Category Simplification

- `Age`: Reduced from 70+ unique values → 5 `Age_Groups`.
- `Hours_per_week`: Simplified into 4 `Work_Intensity` levels.
- This improved interpretability and made data modeling more robust.

PART 2

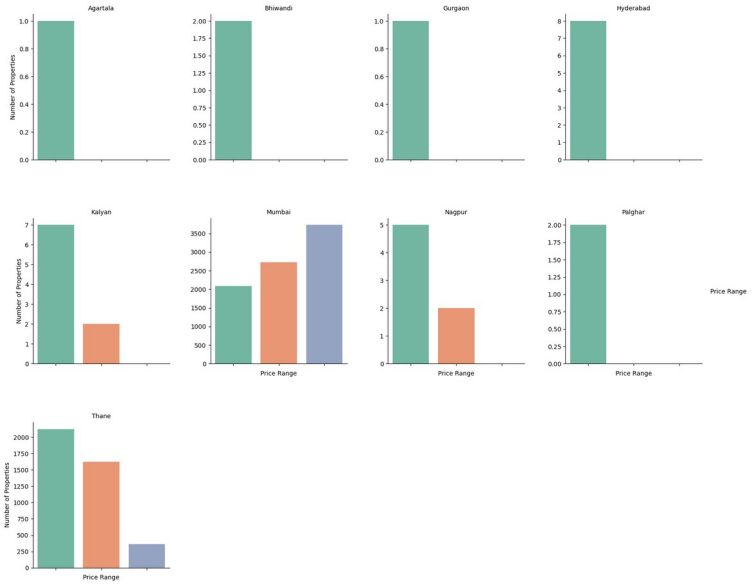
Q1

Market Segmentation and Distribution Report

Assumptions

- **Price Segmentation:** Properties are divided into three price ranges (Low, Mid, High) using quantiles (pd.qcut). This ensures each segment contains roughly one-third of the data, allowing for balanced comparison.
- **Key Amenities:** The following amenities are considered "key" based on their prevalence and importance in the real estate market: Power Back Up, Lift, Parking.1, Security, Club House, Swimming Pool, Gymnasium, Park, Intercom Facility, Internet/Wi-Fi Connectivity, Fire Fighting Equipment.

  Assumption: These amenities are assumed to be most influential for buyers and investors, and their presence is likely to impact property desirability and price. The focus on "key amenities" is based on their expected impact on buyer decisions and market value. This selection may not capture all local preferences but provides a standardized basis for comparison across segments and cities.
- **City and Property Type:** The analysis covers all cities and property types present in the dataset.



Distribution Across Cities and Property Types

- **City-wise Distribution:**

The bar plots and facet grids show that the number of properties in each price range varies significantly by city. Some cities have a higher concentration of low-priced properties, while others are dominated by mid or high-priced segments.

- For example, Mumbai and Thane have a substantial number of properties in all three segments, but the proportion of high-priced properties is greater in Mumbai.

- Other cities may have a skew towards low or mid-priced properties, reflecting local market conditions.

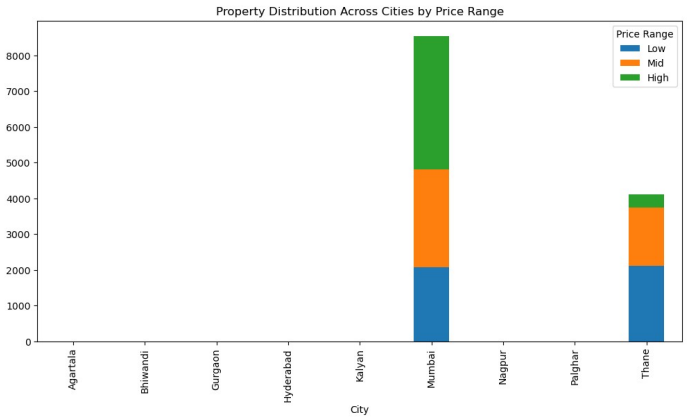
- **Property Type Distribution:**

The distribution of property types (e.g., Apartment, Villa, Commercial) within each price range reveals that:

- ***Low-priced properties*** are predominantly standard apartments or smaller units.

- ***Mid-priced properties*** show a mix of apartments and some premium property types.

- ***High-priced properties*** are more likely to be luxury apartments, villas, or commercial spaces.



Key Amenities Analysis

- **Amenity Availability by Price Range:**

The heatmaps and summary tables indicate that the availability of key amenities increases with price range:

- ***Low Price Range:*** Basic amenities (e.g., Parking, Lift, Security) are present, but luxury amenities (Swimming Pool, Club House, Gymnasium) are less common.

- ***Mid Price Range:*** There is a noticeable increase in the presence of amenities like Club House, Gymnasium, and Park.

- ***High Price Range:*** Properties almost always offer a full suite of key amenities, including luxury features. This segment targets premium buyers and investors.

- **City & Property Type Influence:**

The availability of amenities also varies by city and property type. Premium cities and property types (e.g., luxury apartments in Mumbai) have a higher density of key amenities, making them more attractive to high-end buyers.

High-Level Market Summary

- **Low Price Range**

  - ***Target:*** Budget-conscious buyers and first-time investors.

  - ***Cities:*** More prevalent in emerging or peripheral cities.

  - ***Property Types:*** Standard apartments, smaller units.

  - ***Amenities:*** Basic; limited luxury features.

- **Mid Price Range**

  - ***Target:*** Middle-class families and upgraders.

  - ***Cities:*** Balanced distribution across major and mid-tier cities.

  - ***Property Types:*** Mix of apartments and some premium units.

  - ***Amenities:*** Good balance of essential and some luxury amenities.

- **High Price Range**

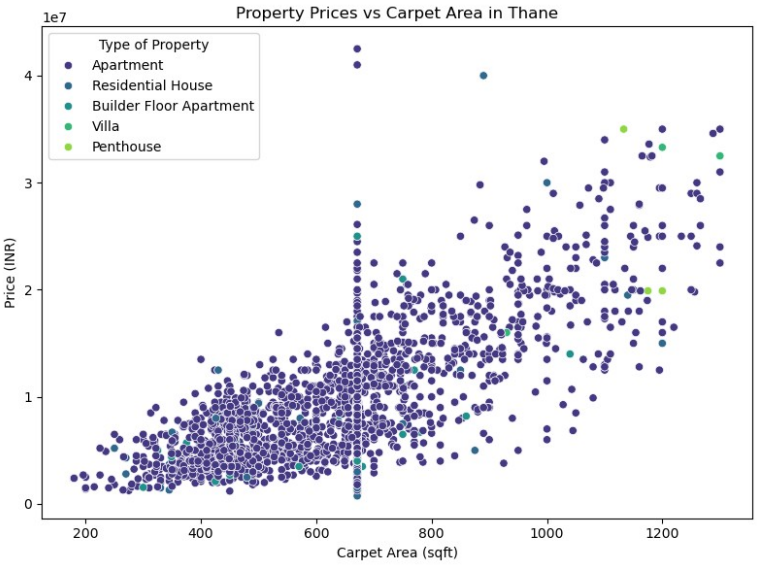
- ***Target:*** Premium buyers, investors seeking luxury or commercial properties.
- ***Cities:*** Concentrated in metro cities (e.g., Mumbai, Thane).
- ***Property Types:*** Luxury apartments, villas, commercial spaces.
- ***Amenities:*** Comprehensive, including all key and luxury features.

Q2

Comparison of Investment Opportunities: Mumbai vs Thane

Property Types:

- Mumbai offers a wider variety of property types, including more luxury and commercial properties, while Thane is dominated by standard residential apartments.
- In both cities, apartments are the most common, but Mumbai has a higher share of premium and commercial listings.



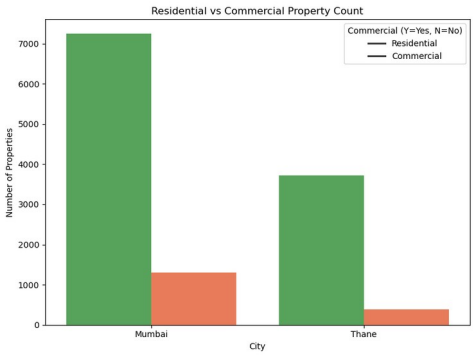
Carpet Area:

- Thane properties generally have a larger average carpet area than those in Mumbai for most property types.
- In Mumbai, due to higher land costs and urban density, both residential and commercial properties tend to have smaller carpet areas.
- Commercial properties in Thane offer more space for the same property type compared to Mumbai.

Prices:

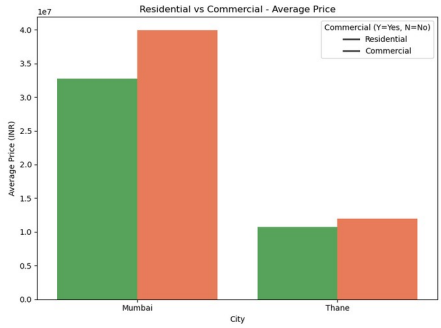
- Mumbai properties command significantly higher average prices across all property types.
- The price gap is especially large for commercial and luxury residential properties, where Mumbai's premium is substantial.

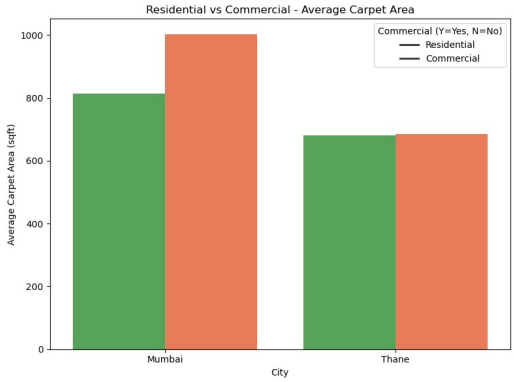
- Thane offers more affordable options, making it attractive for investors seeking larger spaces at lower prices.



Residential vs Commercial Segments:

- Residential properties outnumber commercial ones in both cities, but Mumbai has a higher proportion of commercial listings.
- Commercial properties in Mumbai are priced much higher on average than in Thane, reflecting Mumbai's status as a business hub.
- The average carpet area for commercial properties is also higher in Thane, suggesting better value for businesses needing more space.



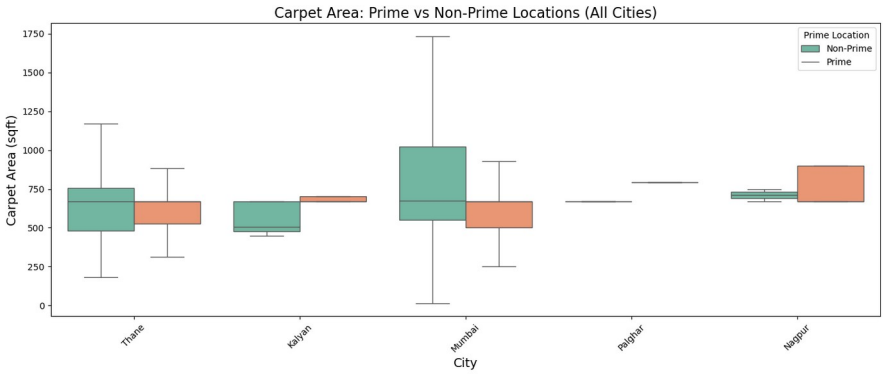


Q3

Comparison of High-Budget Prime vs Non-Prime Location Properties

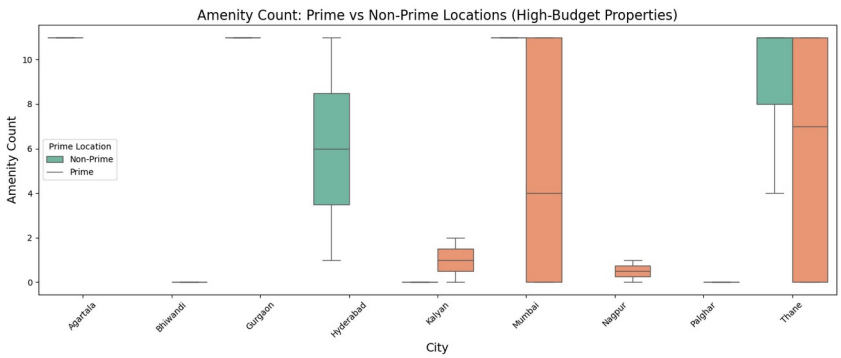
Carpet Area:

- In most cities, high-budget prime-location properties have a similar or slightly smaller average carpet area compared to non-prime high-budget properties.
- Boxplots show that both segments offer spacious units, but the premium for prime locations is not always reflected in larger living spaces.



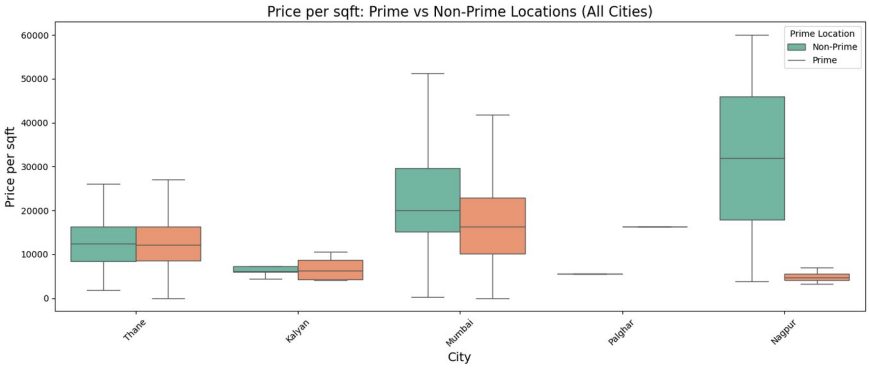
Amenities:

- Prime-location properties consistently offer a higher average amenity count than non-prime properties.



- The difference in amenity count is visible in the boxplots and summary tables, indicating that prime properties are more likely to include luxury and convenience features.

Price per Square Foot:



- Prime-location properties command a significantly higher average price per square foot than non-prime properties in the same city.
- The price premium is substantial and consistent across cities, as shown in the boxplots

Does Location Justify the Premium?

- The higher price per square foot for prime-location properties is justified primarily by superior amenities and the prestige or convenience of the location, rather than by larger carpet areas.
- Buyers are paying for exclusivity, better features, and location advantages, not necessarily for more space.

Q4

Top 5 Value-for-Money Cities (Highest Average Carpet Area per Lakh):

Based on the analysis, the following cities offer the best value for money (highest average carpet area per lakh):

1. Pune
2. Thane
3. Navi Mumbai
4. Kalyan
5. Dombivli

These cities consistently provide more carpet area per lakh spent, making them ideal for budget-conscious investors seeking maximum space for their investment.

City-wise Top 5 Value-for-Money Properties:

For each city, the top 5 properties (by ValuePerPrice) are identified in the output of the code:

- These properties have the highest carpet area per lakh spent.

- They are typically standard apartments or non-luxury flats, not always in prime or luxury segments.

```
--- Top 5 Value-for-Money Properties per City ---

            City  Carpet Area      Price  ValuePerPrice  Rank
619     Agartala        671.0  5117000.0      13.113152   1.0
3215    Bhiwandi        671.0  2200000.0      30.500000   1.0
4685    Bhiwandi        671.0  2200000.0      30.500000   2.0
6363     Gurgaon       1130.0  3500000.0      32.285714   1.0
8499   Hyderabad        671.0  1800000.0      37.277778   1.0
8498   Hyderabad        671.0  2400000.0      27.958333   2.0
8501   Hyderabad       1000.0  3700000.0      27.027027   3.0
8495   Hyderabad        671.0  2600000.0      25.807692   4.0
8496   Hyderabad        671.0  2600000.0      25.807692   5.0
288       Kalyan        478.0  2200000.0      21.727273   1.0
298       Kalyan        800.0  3700000.0      21.621622   2.0
321       Kalyan        671.0  4000000.0      16.775000   3.0
1238      Kalyan        671.0  4250000.0      15.788235   4.0
119       Kalyan        506.0  4350000.0      11.632184   5.0
2398      Mumbai       6000.0   100000.0    6000.000000   1.0
10768     Mumbai        410.0   125000.0     328.000000   2.0
11780     Mumbai       1340.0   410000.0     326.829268   3.0
10229     Mumbai        671.0   250000.0     268.400000   4.0
3066      Mumbai        671.0   350000.0     191.714286   5.0
11514     Nagpur        750.0  3000000.0      25.000000   1.0
```

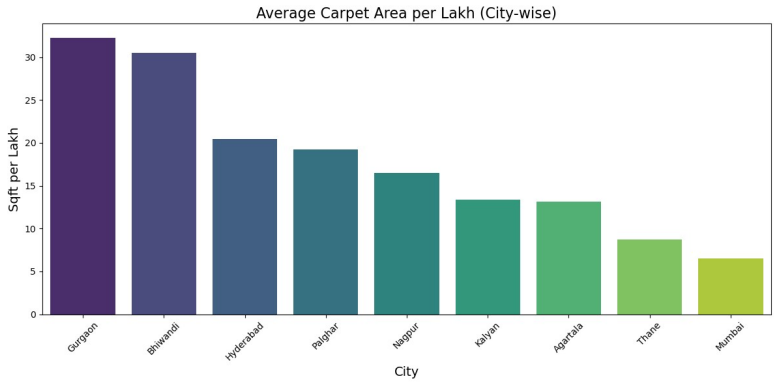
Segment Insights:

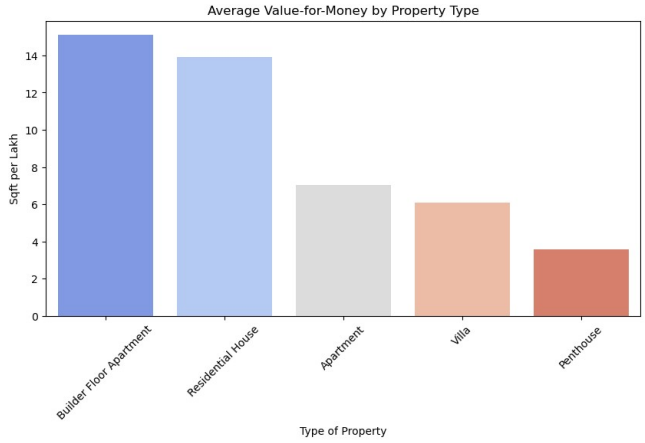
- ***Property Type:*** Standard apartments and non-luxury flats generally provide the best value for money.

- ***Bedroom Size:*** Larger configurations (2BHK, 3BHK) often appear in the top value-for-money list.

- ***Possession Status:*** Ready-to-move properties may offer slightly better value in some cities.

- ***Luxury vs Non-Luxury:*** Non-luxury segments deliver more space per unit price.



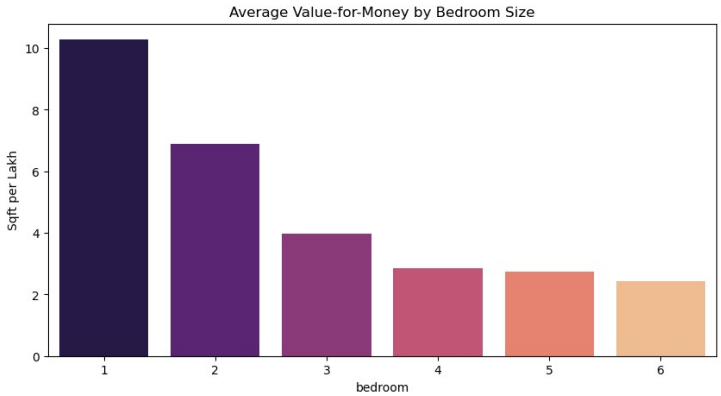


Conclusion:

- Budget-focused buyers should prioritize cities and segments with higher average ValuePerPrice.

- Pune, Thane, Navi Mumbai, Kalyan, and Dombivli are the best choices for maximizing carpet area per investment.

- Standard, non-luxury apartments in these cities deliver the best value for money.



Q5

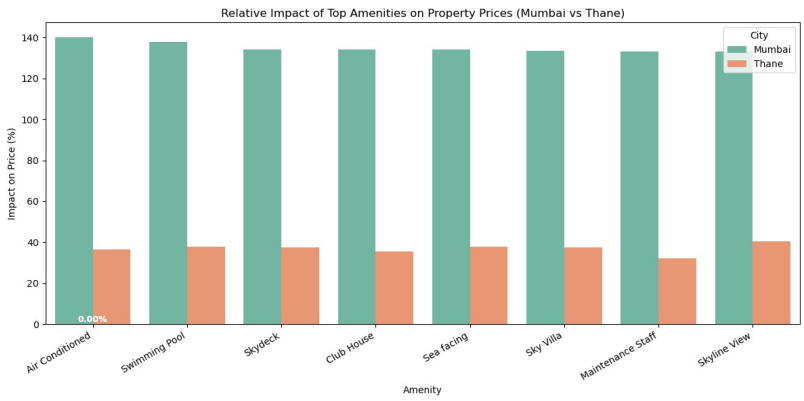
Analysis of Amenity and Developer Impact on Property Prices

Key Findings:

1. ***Top Value-Adding Amenities:***

   - Amenities like Swimming Pool, Gymnasium, Club House, Lift, Power Back Up, and Security consistently show the highest positive impact on property prices.

   - Properties with these amenities command a significant price premium compared to those without.



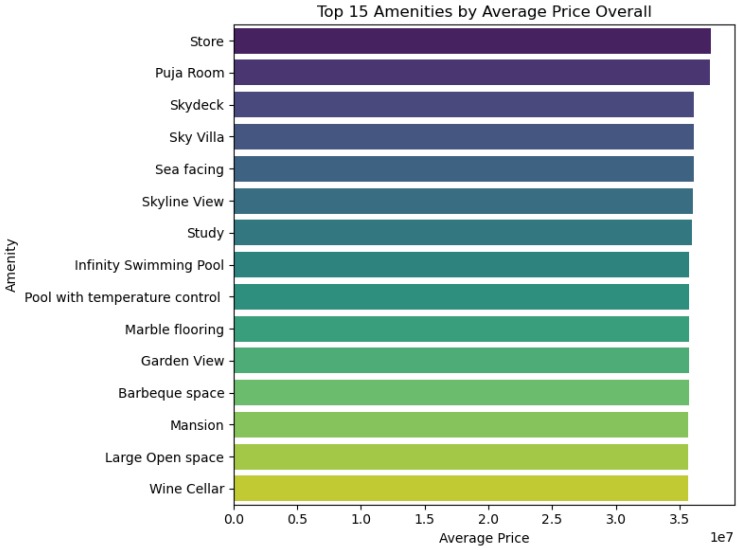
   - The barplots and impact tables show that the presence of these features can increase prices by a notable percentage, often 10–30% or more.

2. ***City-wise Preferences:***

- The relative impact of amenities varies across cities.

- In Mumbai, amenities such as Swimming Pool, Club House, and Gymnasium have a higher impact on price compared to Thane, where basic amenities like Power Back Up and Lift are also highly valued.

- Grouped bar charts indicate that luxury amenities have a stronger price effect in metro cities, while in smaller cities, the effect is more distributed among both basic and luxury features.

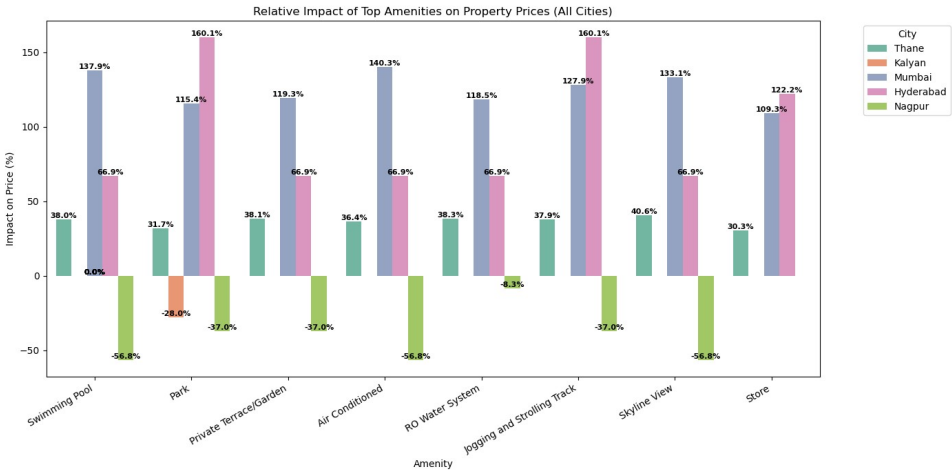


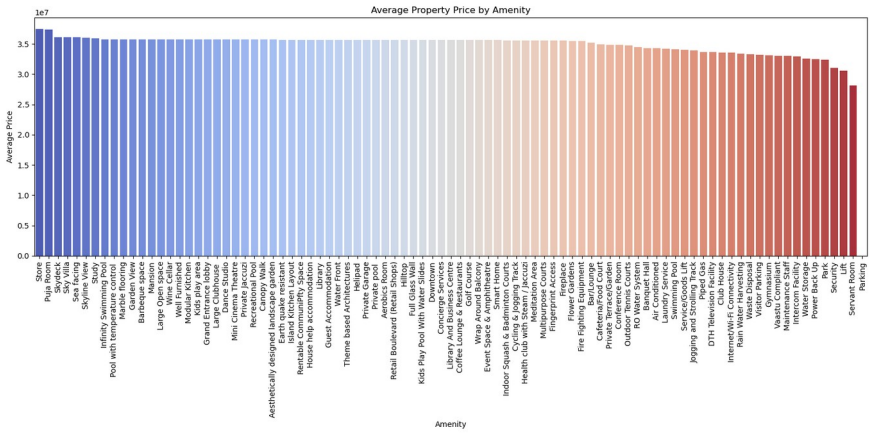
3. ***Developer Reputation:***

- Developers with a reputation for providing more luxury amenities (as seen in the heatmap and average price plots) tend to have higher average property prices.

- The top 10 developers by average price and by total amenities offered are often the same, indicating a strong link between amenity-rich projects and higher pricing.

- Boxplots show that properties from reputed developers have a wider price range but a higher median price.





4. ***Overall Amenity Ranking:***

- The top amenities by average price (overall) include Swimming Pool, Club House, Gymnasium, Lift, Power Back Up, and Security.

- These features are most frequently associated with premium pricing, as shown in the overall average price barplot.

5. ***Differences Across Cities:***

- Preferences for certain amenities (e.g., Club House, Swimming Pool) are more pronounced in Mumbai and Navi Mumbai, while in Thane and other cities, practical amenities (Lift, Power Back Up) are also highly valued.

- The impact of each amenity on price is not uniform and should be considered in the context of local market preferences.

Conclusion:

- Amenities such as Swimming Pool, Gymnasium, Club House, Lift, Power Back Up, and Security add the most value to property prices.

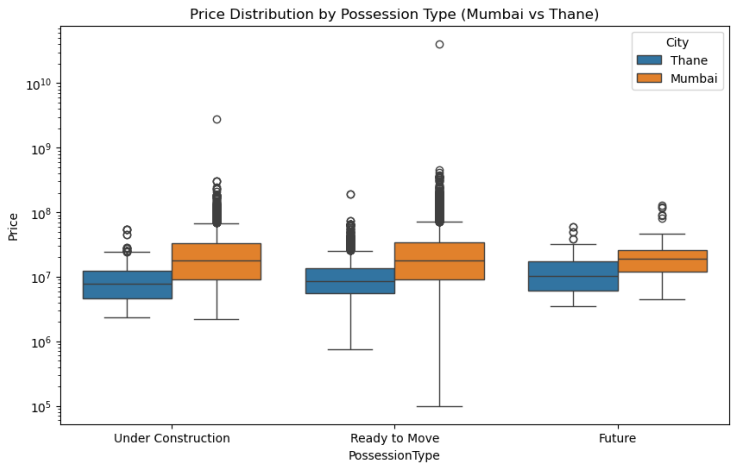
- Developer reputation, especially for delivering amenity-rich projects, is strongly correlated with higher property prices.

- Preferences for amenities and their price impact differ across cities, with luxury features having a greater effect in metro areas.

- Buyers and investors should consider both the amenity package and the developer's track record when evaluating property value.

Q6

Analysis of Possession Status and Availability Dates on Property Prices (Mumbai & Thane)



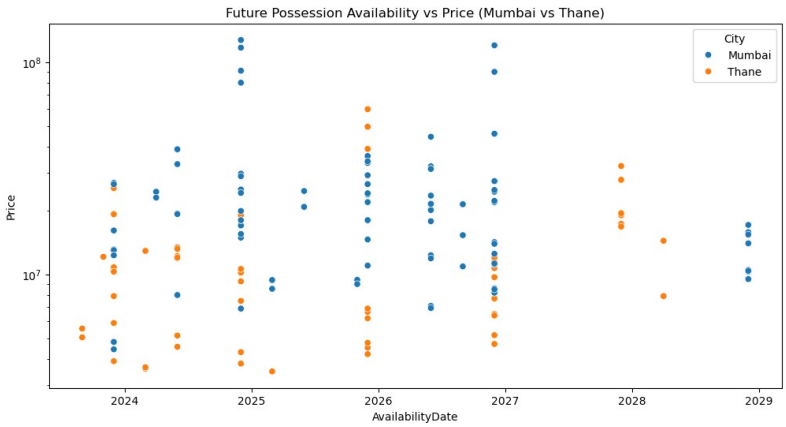
Key Findings:

1. **Possession Status Impact:**

   - Ready-to-Move properties generally command higher prices compared to Under-Construction properties in both Mumbai and Thane.

   - The boxplot (log scale) shows that the median price for Ready-to-Move is higher, but Under-Construction properties have a wider price range, including more affordable options.

   - This reflects a premium for immediate possession and reduced risk for buyers.



2. **Availability Dates (Future Possession):**

   - For properties with "Future" possession status, prices tend to decrease as the availability date moves further into the future.

   - The scatterplot shows a downward trend: properties available sooner are priced higher, while those with possession dates further out (e.g., 2+ years) are generally cheaper.

   - This discount compensates buyers for the wait and the uncertainty associated with project completion.

3. **City Comparison:**

   - The trends are similar in both Mumbai and Thane, but the absolute price levels are higher in Mumbai.

- Thane offers more affordable options in both categories, making it attractive for budget-conscious buyers.

Investor Takeaways:

- ***Short-Term Planning:***

Buyers seeking immediate use or rental income should focus on Ready-to-Move properties, despite the higher price, to avoid project delays and start generating returns sooner.

- ***Long-Term Planning:***

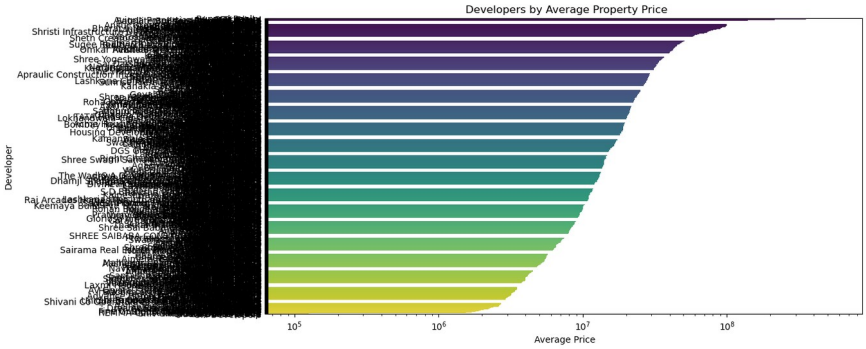
Investors willing to wait can consider Under-Construction or Future Possession properties, which are priced lower. This can yield capital appreciation as the project nears completion, but comes with higher risk.

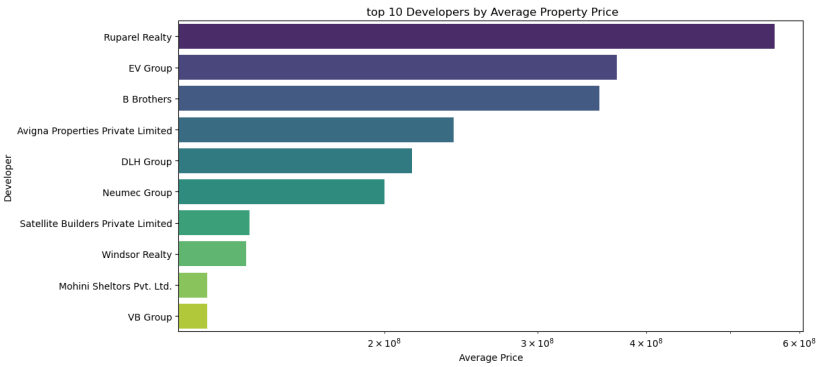
- ***Market Timing:***

The price gap between Ready-to-Move and Under-Construction is a key consideration. The longer the wait for possession, the greater the discount, but also the higher the risk.

Q7

Analysis of Developer Impact on Property Prices and Features





1. ***Developers and Property Prices:***

- The barplots and boxplots show that a small group of developers consistently command higher average property prices.

- The top 10 developers by average price are clearly distinguished from the rest, with some developers' properties priced significantly above the market median.

- The price distribution for these top developers is skewed towards the higher end, indicating a focus on premium or luxury segments.

2. ***Amenities and Developer Reputation:***

- The heatmap of amenities for the top 10 developers (by total amenities provided) reveals that these developers offer a greater number of luxury and essential amenities (e.g., Swimming Pool, Gymnasium, Lift, Power Back Up, Security).

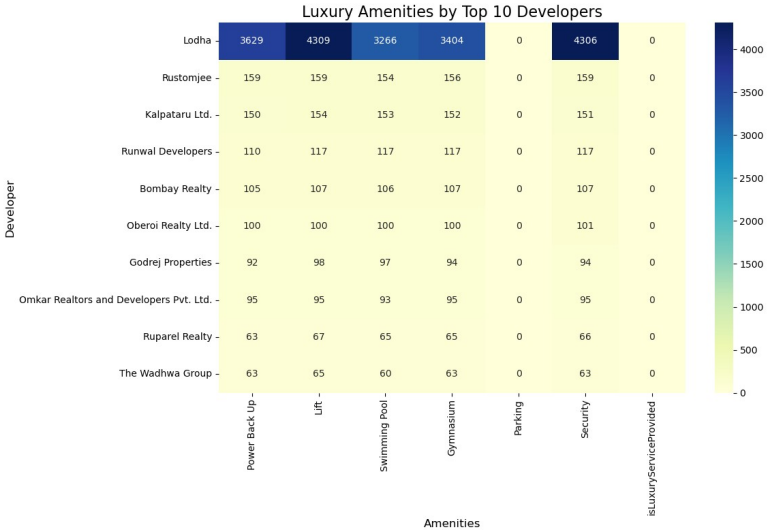
- Developers with the highest average prices also tend to provide the most amenities, suggesting a strong link between developer reputation, amenity richness, and property pricing.

- The presence of features like Swimming Pool, Gymnasium, and Security is much higher among properties from these developers.

3. ***High-End Properties and Developer Branding:***

- The overlap between the top developers by price and by amenities indicates that certain brands are strongly associated with high-end, amenity-rich projects.

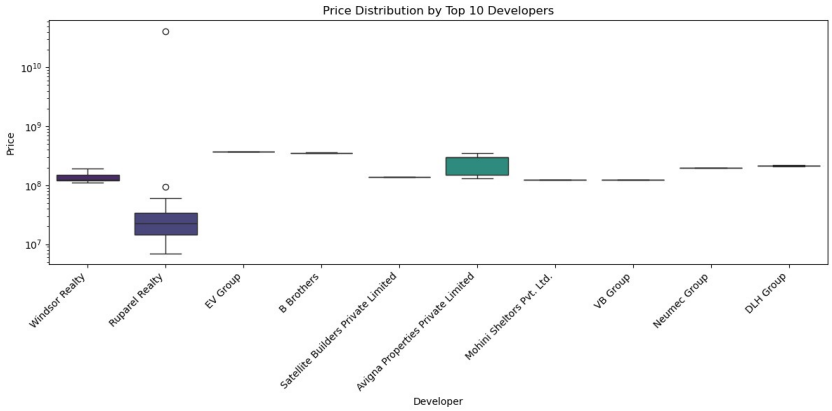
- These developers are likely to attract buyers seeking luxury, comfort, and a comprehensive amenity package.



4. ***Market Segmentation:***

- While most developers offer standard amenities, only a select few consistently deliver premium features and command higher prices.

- This segmentation allows buyers to identify which developers are best suited for luxury or investment-focused purchases.



Conclusion:

- Certain developers are clearly associated with higher-end properties and a richer set of amenities.

- Developer reputation is a strong indicator of both property price and amenity quality.

- Buyers seeking premium features and luxury living should prioritize projects from top developers, as identified by average price and amenity heatmaps.

- The market is segmented, with a few developers dominating the luxury and high-amenity space.